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PALEOMAGNETIC EVIDENCE FOR POST-LATE MIOCENE INTRA-ARC ROTATION OF SOUTH KYUSHU, JAPAN

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Abstract. Paleomagnetism of sedimentary rocks of the late Miocene Uchiumigawa Group exposed in southeast Kyushu yielded 17 tilt-corrected paleomagnetic directions with significant westward declinations. Fourteen of these were of reversed polarity, and three at the intervening horizons were of antipodal normal polarity. The overall formation mean direction is D = 322.0°, I = 48.6° with α_{95} = 6.0°. This is statistically indistinguishable from the mean direction of D = 331.2°, I = 41.3° with α_{95} = 9.9° for the middle Miocene deposits in Tanegashima Island of the northern Ryukyu arc. The common mean direction is D = 333.2°, I = 45.1° with α_{95} = 4.9°. This indicates that both south Kyushu and the northernmost Ryukyu arc have experienced 27° ± 6° of counterclockwise rotation with respect to the Eurasian continent after the latest Miocene, or during the last 6 m.y. This counterclockwise rotation cannot be fully explained by either a conventional model which links an arc rotation with the formation of back-arc oceanic crust, or the collision of the Kyushu-Palau ridge with southwest Japan arc. We propose that this intra-arc rotation took place during the period of extension of the continental crust behind the north Ryukyu arc. This study demonstrates that arc rotation may generally occur when the back-arc spreading is of a much less advanced phase.

INTRODUCTION

On the basis of paleomagnetic study, Kawai et al. [1961] first interpreted the bent framework of the Japanese arc (Figure 1) as a consequence of a counterclockwise (CCW) rotation of northeast Japan and a clockwise (CW) rotation of southwest Japan. Since then numerous paleomagnetic studies have been devoted to clarify the time and the amount of the geotectonic motion. Recent increase in well-dated paleomagnetic data [e.g., Otofuji et al., 1991] has demonstrated that southwest Japan rotated CW by 40°-50° at the early middle Miocene. Northeast Japan experienced a contemporaneous CCW rotation [e.g., Tosha and Hamano, 1988], but the amount and the duration of rotation are not so well documented. Because of the synchronism of the two different rotations, Otofuji and Matsuda [1987] proposed that the bend of the Japanese arc was caused by a rapid, fan-shaped back-arc spreading of the northeastern and southwestern portions of the Japan Sea in the middle Miocene. Also, because of the consistent paleomagnetic directions in their study areas, they favored that both southwest and northeast Japan rotated in a single stage, as two independent, rigid blocks.

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However, not all paleomagnetic studies support this rigid body rotation model. Itoh [1988] found that pre-Miocene rocks in the eastern part of southwest Japan displayed an easterly paleomagnetic deflection significantly smaller than that reported from the western portions. Hirooka et al. [1986] obtained a similar result from north central Japan. It thus seems more realistic to assume that the eastern part of southwest Japan rotated less CW than the western part. Alternatively, central Japan may consist of a set of small blocks, each being rotated in a different manner from the western main body [e.g., Kanaori et al., 1990]. According to Hyodo and Niitsuma [1986], an early Miocene terrane in south central Japan undertook a CW rotation in excess of 90°. They interpreted the anomalously large rotation as a consequence of the collision of the Izu-Bonin arc with central Japan since Miocene time [e.g., Matsuda, 1978]. Up to 50° CW deflections in declination were determined on the Pleistocene strata exposed in the northeast Izu Peninsula [Koyama and Kitazato, 1989]. This intra-arc deformation can also be explained by the collision of the Izu block with central Japan which postdates the Japan Sea opening.

From northeast Japan, although the majority of data are characterized by westerly deflection in declination, several paleomagnetic data with easterly declinations were obtained by Otofuji et al. [1985]. These inconsistent directions were observed in a limited area along the Tanakura shear zone (TSZ in Figure 1). Thus they proposed that the anomalous paleomagnetic directions reflected a local, CW rotation of a small block, caused by a right-lateral displacement along the nearby transcurrent fault.

On the basis of analyses of on-land geologic structures, Kanaori [1990] has proposed that not only central Japan but southwest Japan can be divided into more than 10 subblocks, each bounded by strike-slip faults, and that these unit blocks were simultaneously rotated by the opening of the Japan Sea. Kano et al. [1990] concluded that major folds in southwest Japan had been formed during the middle Miocene, contemporaneous with the CW rotation of southwest Japan. Likewise, Faure and Lalevée [1987] stressed that the bent trend of the pre-Miocene structures of Japan had been produced by a broad-scale, ductile flexural-slip fold, as a result of the Miocene opening of the Japan Sea. On the basis of a right-lateral pull-apart opening model of the Japan Sea [Lallemand and Jolivet, 1985] and a laboratory analog experiment, Jolivet et al. [1991] concluded that the Japanese arc consists of several subblocks, which rotated CW in the south and CCW in the north.

Thus it seems more plausible to assume that the present structural framework of the Japanese arc was established principally in Miocene time, accompanied with ductile intra-arc deformation and with tectonic rotation of smaller subblocks. In particular, central Japan has been more strongly distorted by compression produced by the Pliocene-Quaternary collision of the Izu-Bonin ridge.

We present results of a paleomagnetic study on the upper Miocene sedimentary rocks exposed along the southeastern coast of Kyushu. Combining the new paleomagnetic data with the previous ones from the middle Miocene deposits in Tanegashima Island off south Kyushu [Kodama et al. 1991], we obtained middle to late Miocene paleomagnetic directions for south Kyushu. We then propose a model for the tectonic history of south Kyushu from the middle Miocene to the present. Our model supports that Kyushu is not a single tectonic block but a composite of north and south subunits.

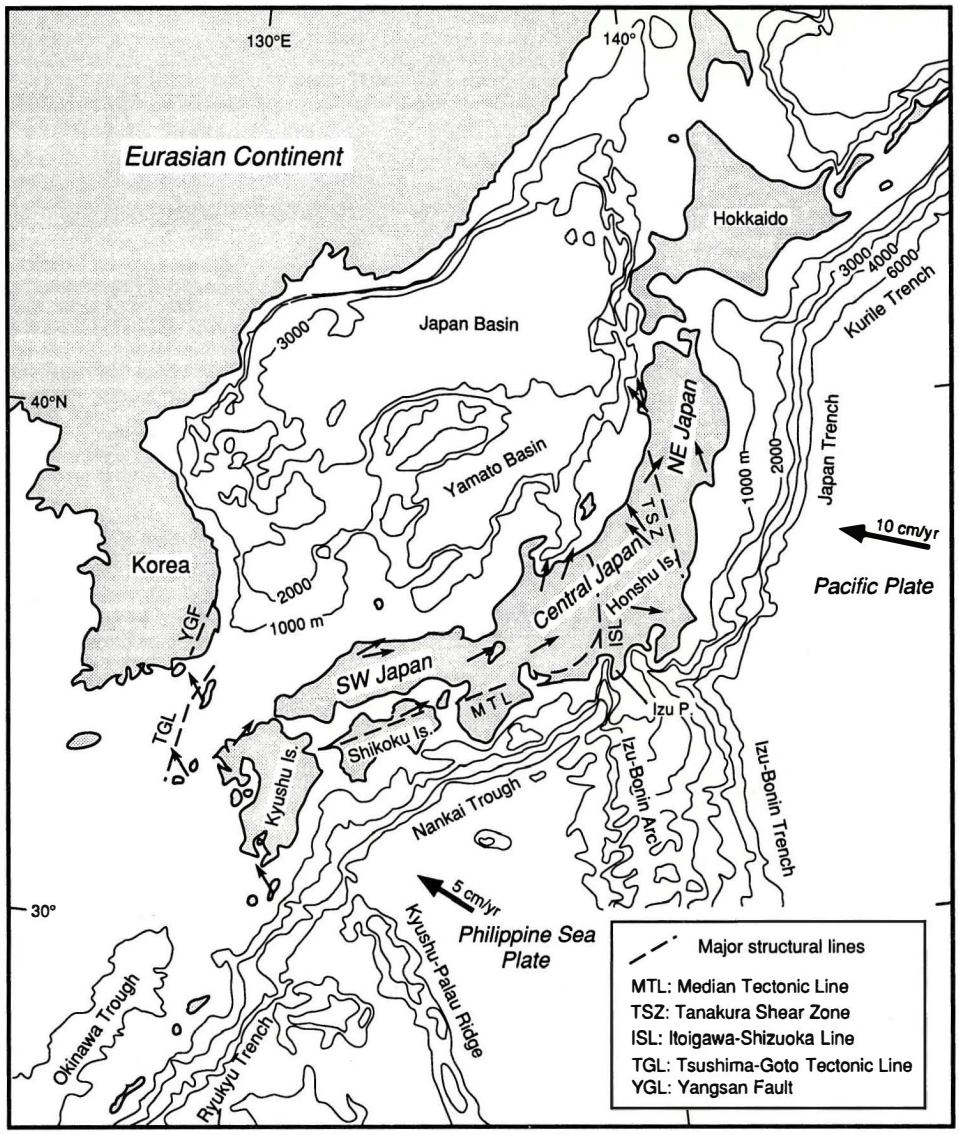


Fig. 1. General morphology of the Japanese arc and its surrounding area, with major tectonic lineations. Small arrows denote the representative paleomagnetic declinations for northeast Japan [Otofuji et al., 1985; Tosha and Hamano, 1988], central Japan [Hirooka et al., 1986; Hyodo and Niitsuma, 1986; Itoh, 1988] and southwest Japan [Otofuji and Matsuda, 1987; Hayashida, 1986] including Shikoku [Kodama, 1989], Kyushu [Ishikawa, 1990], and nearby islands [Kodama et al., 1991; Ishikawa and Tagami, 1991]. Thick arrows are present motions of the Philippine Sea plate and the Pacific plate relative to the Eurasian plate [American Association of Petroleum Geologists, 1981].

PREVIOUS PALEOMAGNETIC DATA FROM KYUSHU

The CW rotation of southwest Japan was proposed on the basis of extensive sampling of Miocene tuffs [e.g., Otofuji and Matsuda, 1987] and sedimentary rocks [Hayashida, 1986], distributed widely in the western part of Honshu Island. Paleomagnetic data from the other areas in southwest Japan, including Kyushu and Shikoku Islands, are still sparse. The representative paleomagnetic directions for Kyushu and its surroundings are shown in Figure 2. Remanence directions of the Cretaceous sedimentary rocks in Shikoku Island (9 in Figure 2) exhibit eastward declinations in the same manner as the Miocene rocks in Honshu (e.g., 8 in Figure 2). A similar CW deflection in declination has been reported by paleomagnetism of the Late Cretaceous granite (6) and the Paleogene sedimentary rocks (5) in north Kyushu. In contrast, middle Miocene igneous rocks in the Tsushima (4) and Goto Islands (3) off north Kyushu have given westward magnetic directions. These directions have been interpreted to represent local, CCW block rotations of these islands due to sinistral displacement of the submarine Tsushima-Goto tectonic line (TGL in Figures 1 and 2) [Ishikawa and Tagami, 1991]. Several middle Miocene granites in south Kyushu (2) yielded components deflected CCW, while those in central eastern

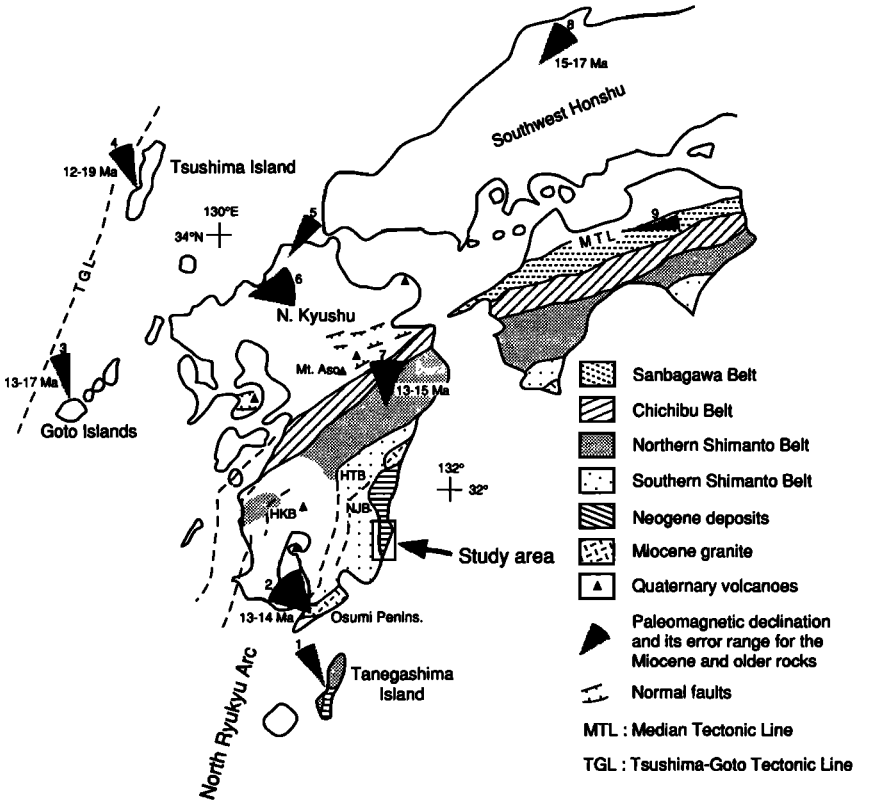


Fig. 2. General geology south of the Median Tectonic Line in Kyushu and Shikoku islands. Study area is located in the south of Miyazaki Prefecture. The paleomagnetic declinations with errors of 95% confidence limit are from late Miocene deposits (1, Tanegashima Island [Kodama et al., 1991]), middle Miocene granitic rocks (2, Osumi Peninsula [Ishikawa and Torii, 1986]; 3, Goto Islands, and 4, Tsushima Island [Ishikawa and Tagami, 1991]; 7, Okueyama [Torii and Ishikawa, 1986]), middle Miocene volcanic rocks (8, southwest Honshu [Otofuji et al., 1991]), Oligocene deposits (5, north Kyushu [Ishikawa, 1990]); Cretaceous granitic rocks (6, north Kyushu [Ito and Tokieda, 1986]), and Cretaceous deposits (9, Shikoku [Kodama, 1989]). Abbreviations are HKB, Hokusatsu Bend [Hashimoto, 1962]; HTB, Hitoyoshi Bend [Murata, 1987]; and NJB, Nojiri Bend [Murata, 1987].

Kyushu (7) possessed components directed parallel to the present north. However, there is controversy concerning tilt correction on these Cretaceous-Miocene paleomagnetic data from Kyushu, because most of them were obtained from intrusive rocks.

Voluminous volcanic material of Pliocene-Pleistocene age overlies pre-Tertiary basement in central Kyushu. Combined K-Ar age determinations and preliminary paleomagnetic measurements [e.g., Suto, 1985] (not in Figure 2) revealed remanence directions with both normal and reversed polarities, correlatable to polarity chrons and subchrons during the last 4 m.y. Diversity in site mean directions is generally large due to incomplete demagnetization treatment, secular variation, or scarcity of sample numbers per site. Thus no systematic difference in declination from the present geomagnetic field has been recognized in this region.

Kodama et al. [1991] carried out a magnetostratigraphic study on the middle Miocene sedimentary rocks exposed in Tanegashima Island (1) off south Kyushu. Because of the consistent, westward declination deflection throughout the formations, they concluded that Tanegashima Island had rotated CCW about 30° relative to southwest Japan during the last 10 m.y. They further predicted that not only Tanegashima Island but south Kyushu could have suffered the CCW rotation, because of their similarity in geological structure. They interpreted the rotation as an intra-arc deformation which had been produced by collision or buoyant subduction of the Kyushu-Palau ridge (see Figure 1). However, due to the lack of exposures and the limited occurrence of microfossils for age determinations, it was hard to date the CCW rotation more precisely than during the last 10 m.y.

The purpose of this paleomagnetic study is to obtain better age constraints for the predicted CCW rotation of south Kyushu and to clarify an areal extent of the rotated region. This is why we examined the upper Miocene Uchiumigawa Group in southwest Kyushu, which exhibits a structural trend similar to that of the middle Miocene formations in Tanegashima Island.

GEOLOGIC SETTING

The pre-Miocene formations in Shikoku Island exhibit a clear zonal distribution with an ENE-WSW strike (Figure 2). These formations extend westward into south Kyushu, but at

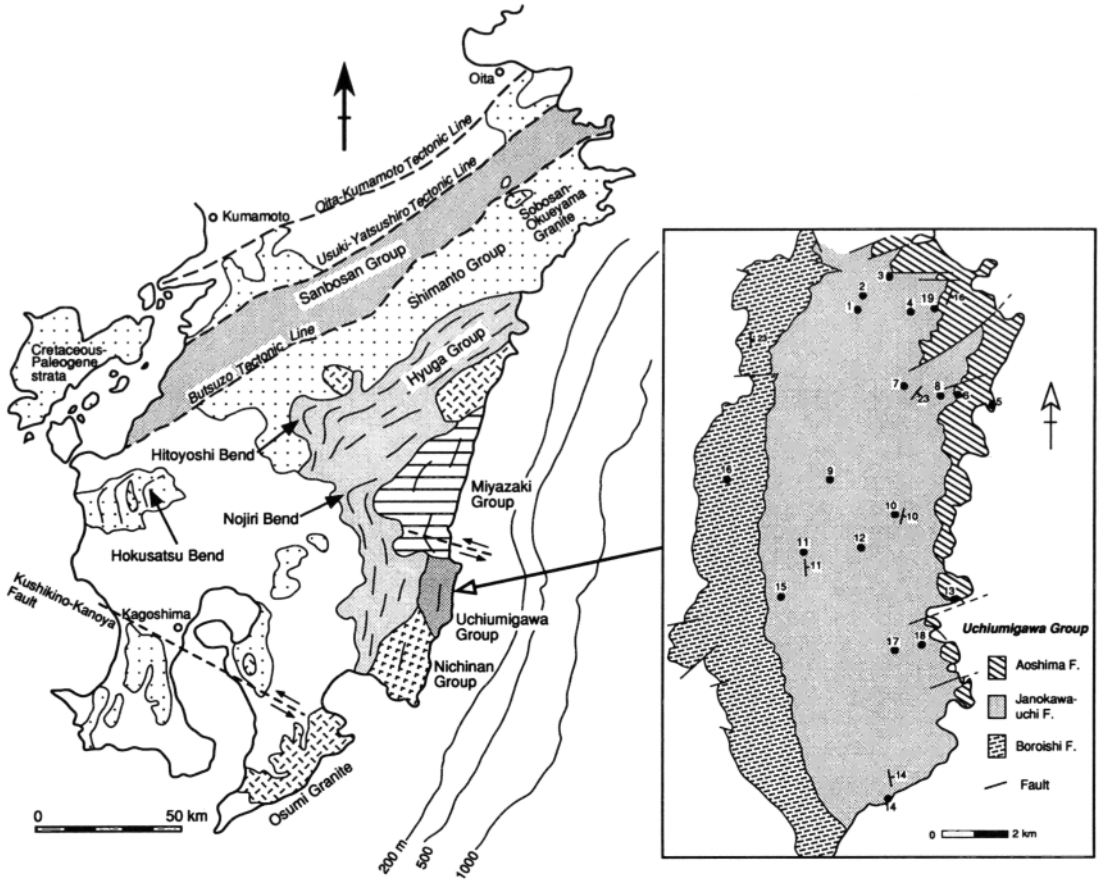


Fig. 3. Geologic map south of the Oita-Kumamoto Tectonic Line that is the western continuation of Median Tectonic Line into central Kyushu (reproduced partly after Imai et al. [1975] and Murata [1987]), with site locations (solid circles with site numbers) in this study of the Uchiumigawa Group. The map of the group is after H. Nakagawa (personal communication, 1991).

its southwestern part the trend digresses more southward and continues to the north Ryukyu arc. Thick Tertiary sedimentary rocks are well exposed with a N-S trend in southeast Kyushu (Figure 3). They are divisible into three groups: the Paleogene and lower Miocene Nichinan Group [Kato, 1985], the upper Miocene Uchiumigawa Group [Takehi, 1978], and the late upper Miocene to the upper Pliocene Miyazaki Group [Suzuki, 1987]. These stratigraphic groups are separated by unconformities (Figure 4). The Uchiumigawa and the overlying Miyazaki groups consist mainly of alternations of sandstone and siltstone with a homoclinal structure, gently dipping to the east about a N-S strike [Suzuki, 1987]. In contrast, the Nichinan Group is an assemblage of dismembered sandstone and siltstone layers with complex geologic structure [Kato, 1985; Nakagawa et al., 1988].

According to Takehi [1978], the Uchiumigawa Group is subdivisible into three lithostratigraphic units; the Boro-ishi, Janokawauchi, and Aoshima formations, in ascending order (Figure 4). The layers dip 20°-40° eastward about a N-S strike. The total thickness is approximately 2000 m. The Boro-ishi Formation is composed of thick sandstone layers, with abundant intercalations of conglomerate in the lower part. The Janokawauchi Formation consists of alternations of sandstone and shale, and the Aoshima Formation is predominantly composed of siltstone. Planktonic foraminifera assemblages from these formations [Takehi, 1978; H. Nakagawa, personal communication, 1991] indicate that the Boro-ishi Formation corresponds to the N15 zone of Blow [1969], the Janokawauchi Formation to N16, and the Aoshima Formation to N17. This means that the age of the Uchiumigawa Group ranges from earliest late Miocene to late late Miocene (10.5 to 6 Ma, according to Berggren et al. [1985]).

SAMPLING AND LABORATORY PROCEDURES

We collected samples at 19 localities of the Uchiumigawa Group, each belonging to a different stratigraphic level: three from the Aoshima Formation, 15 from the Janokawauchi Formation, and one from the Boro-ishi Formation (Figure 3). These samples were taken from either the finer-grained portions of sandstone layers or the siltstone and interbedded mudstone layers.

All the samples were taken using a gas-powered drill in the field, with orientation by a magnetic and/or a sun compass. Five to eight core samples, with one to three cores per layer, were drilled at individual localities. The 2.54 cm (1 inch) diameter field cores were sectioned into 22 mm long specimens. In each core sample, two specimens from the

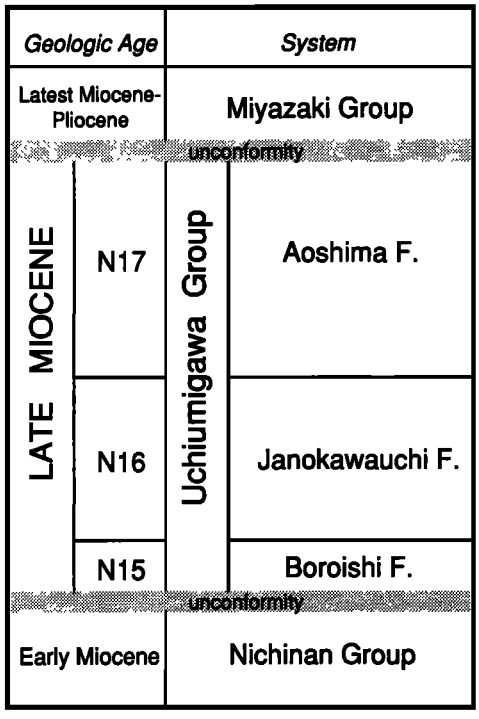


Fig. 4. Stratigraphy of the Miocene strata in southeast Kyushu, partly after Kato [1985] and Suzuki [1987]. Biostratigraphy for the Uchiumigawa Group as N15 to N17 in Blow's zonation is after Takehi [1978] and H. Nakagawa (personal communication, 1991).

deepest portion, or one if not long enough, were used for measurements. A total of 215 specimens were collected from the Uchiumigawa Group and subjected to paleomagnetic analyses. Bedding attitude at each site was measured on at least five bedding surfaces and averaged. This averaging produced an error of less than 3°.

Magnetization measurements were performed with a three-axis 2G cryogenic magnetometer (2G model 760R) equipped with an automated sample handler, using five orientations per specimen. The noise level is 3×10^{-6} to 5×10^{-6} A/m in total magnetic moment. As a pilot study to determine the effective demagnetization method for routine measurements, at least one pair of specimens was prepared at individual sites, and each companion specimen was subjected to progressive alternating-field (af) or thermal demagnetization treatment. The thermal treatment was made in air in a laboratory-made oven shielded by a three-layered μ-metal, in which Earth magnetic field is reduced lower than 10 nT. The progressive thermal demagnetization was made in steps from 100°C with 25° increments until the remanence intensity became too weak for meaningful measurement. Bulk magnetic susceptibility was measured after each cooling run with a Bartington susceptibility meter (model M.S.2) in order to monitor thermal alteration of magnetic minerals likely to occur at higher temperatures. The thermal demagnetization was also halted when an increase in either remanence intensity or magnetic susceptibility greater than 1 order of magnitude was encountered. The af demagnetization was conducted on a laboratory-produced demagnetizer consisting of a three-axis tumbler, alternating current oscillator and a 500 W audio-amplifier, capable of producing a maximum field intensity of 45 mT. The progressive af demagnetization was taken in steps from 2.5 to 25 mT with 2.5 mT increments, and from 30 to 45 mT with 5 mT increments. If the af demagnetization was ineffective in isolation of characteristic component of magnetization, thermal demagnetizations at 100°C to 350°C were carried out.

Initial and demagnetized remanence vectors of each specimen were projected on an orthogonal diagram [Zijderveld, 1967]. The characteristic components were isolated by applying the principal component analysis of Kirschvink [1980] to those sets of vectors which displayed a linear trend on the diagram. The components defined by a minimum of four vector endpoints and, if necessary, the origin, with a MAD (maximum angular deviation of Kirschvink [1980]) smaller than 15°, were employed for further data analyses. When multiple components with overlapping magnetic coercivity or blocking temperature spectrum were recognized, the poles to their demagnetization planes were calculated the same way. If both demagnetization circle and direct observational data were available at a single site, the combined analysis of the direct and indirect data by McFadden and McElhinny [1988] was applied to estimate the common direction and its statistical confidence.

PALEOMAGNETIC RESULTS

As a pilot study, 21 pairs of companion specimens were selected from representative cores of different sites and lithologies. They were divided into halves, and each group was subjected to progressive af or thermal demagnetization. Stable characteristic components with low to moderate median destructive field (MDF) intensities between 5 and 15 mT were detected from 13 of the af demagnetized group. These components are directed almost parallel to the present field, and no higher-coercivity component could be isolated upon steps between 25 and 40 mT, during which they exhibited unstable magnetic directions. Only three specimens possessed characteristic high-coercivity components, and the others showed extremely erratic behavior. All the thermally demagnetized pilots had low-temperature components with unblocking temperature lower than 200°C. Superposed high-temperature components could be isolated from seven specimens. Two of these components gave directions identical to those of the high-coercivity components detected from the af demagnetized companions. Therefore the remaining 194 specimens were all subjected to stepwise thermal demagnetization. Excessive increases in both bulk magnetic susceptibility and magnetization intensity were frequently observed after heating above 350°C. These increases suggest that magnetite was produced by oxidization of reduced phases of iron like pyrite and pyrrhotite. Examples of the demagnetization diagrams are shown in Figure 5.

Intensities of initial remanent magnetization were generally weak, between 2.85×10^{-6} and 2.69×10^{-8} kA/m (1 kA/m = 1 emu/cc), with the majority (80%) ranging from 1.1×10^{-7} to 1.0×10^{-6} kA/m. Stepwise thermal demagnetization revealed that most of the specimens carried those components which could be removed during early steps at temperatures lower than 200°C. Examination of the vector endpoints of these soft components on orthogonal plots enabled us to isolate a total of 159 directions. Each component was calculated using at least

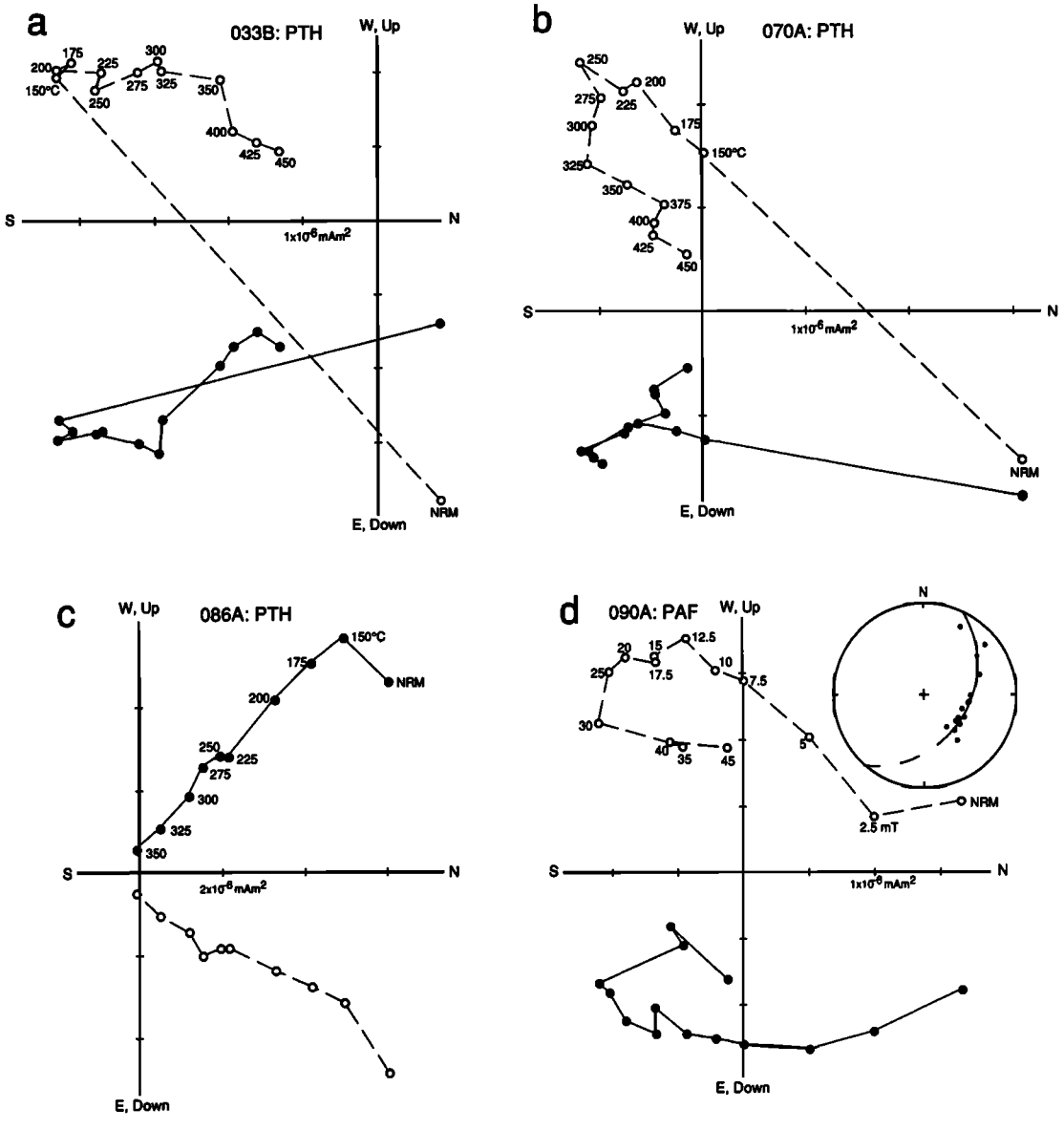


Fig. 5. Examples of demagnetization diagrams. Solid and open circles are projections onto the horizontal and the vertical N-S plane, respectively, in geographic coordinates. PTH is progressive thermal demagnetization, and PAF is progressive alternating field demagnetization. Unit is mAm^2 (1 mAm^2 = 1 emu) in bulk magnetic moment. (a), Site 5 in the Aoshima Formation of the Uchiumigawa Group. Reversed polarity with characteristic direction, D = 136.2°, I = –29.5°, maximum angular deviation (MAD) = 14.0° computed by least squares line fit using steps at temperatures above 300°C. Low-temperature component with direction, D = 344.4°, I = 47.7°, MAD = 2.3° from natural remanent magnetization (NRM) through 200° steps is likely a present field overprint. (b), Site 10 in the Janokawauchi Formation. Reversed polarity with D = 138.9°, I = –37.0°, MAD = 12.3° from steps above 325°. (c), Site 13 in the uppermost Janokawauchi Formation. Normal polarity with D = 313.9°, I = 18.9°, MAD = 4.3° from steps above 150°. (d), Site 14 in the Janokawauchi Formation, with a curved demagnetization trajectory from NRM through 25 mT, indicating two components with overlapping high and low magnetic coercivities. No characteristic component can be isolated due to the erratic behavior at steps above 25 mT. The equal-area projection (upper hemisphere) shows the demagnetization great circle, with its pole directing D = 116.1°, I = 39.4°, MAD = 1.5°, obtained by least squares plane fit using NRM through 25 mT steps.

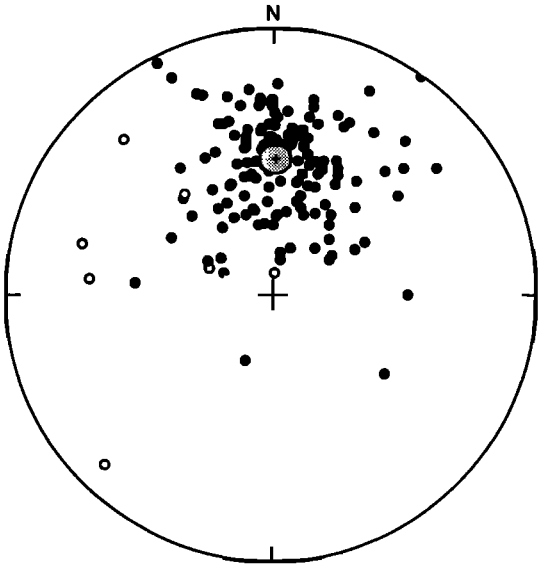


Fig. 6. Directions of all the magnetization components with low unblocking spectra isolated with least squares analyses; each computed using vector endpoints displaying univectorial trend during early steps of thermal demagnetization at temperatures lower than 200°C. Equal-area projection in geographic coordinates, with solid circles on the lower hemisphere. The cross within the oval denotes the mean direction with 95% confidence error.

four endpoints with MAD smaller than 15°. All of these directions are plotted in Figure 6, indicating the concentration around the dipole field direction expected at the latitude of the study area (I = 51°). The overall mean in geographic coordinate was D = 1.3°, I = 47.8° with α_{95} = 4.4°, which is statistically indistinguishable from the predicted dipole field direction. Upon structural correction at each site, the mean direction shifted slightly eastward, that is, D = 17.2°, I = 47.0° with α_{95} = 4.5°, but no better grouping of statistical significance was achieved. This seems to imply that these components are secondary overprints, probably viscous remanent magnetizations acquired after tectonic deformation.

Of the 215 specimens demagnetized, characteristic magnetization components with higher unblocking temperatures were isolated for 88 specimens. Many of these components are directed up and to the southeast in both geographic and stratigraphic coordinates. Those from sites 11 and 12 in the Janokawauchi Formation and from site 13 in the Aoshima Formation were downward in the northwest quadrant, being antipodal to the others. No useful line-fit direction was discernible in the remaining specimens because of either too weak intensity or erratic behavior at higher demagnetization steps. However, demagnetization great circles were defined for the 71 specimens, in which no higher temperature component could be computed but curved demagnetization trajectories were recognizable up to 250°-300°C steps (Figure 5d). These great circles at individual sites appear to converge on the cluster of directions obtained by direct line-fit analysis (Figure 7). Accordingly, using the demagnetization plane data and the direct line-fit data, 16 site mean directions (sites 2-4 and 6-18) were estimated by the method of McFadden and McElhinny [1988]. One site mean direction from site 5 was computed

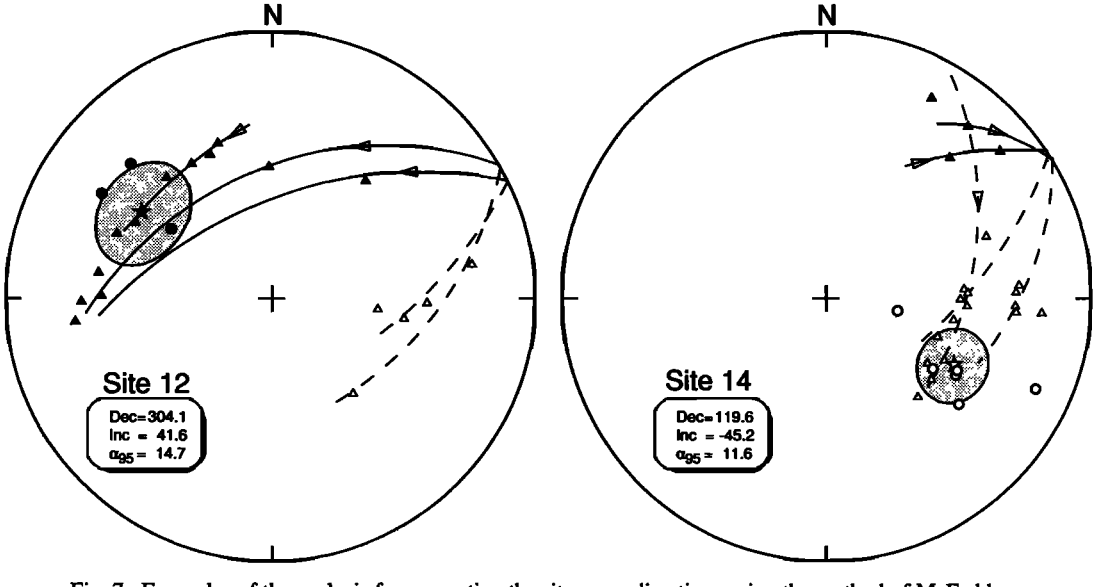


Fig. 7. Examples of the analysis for computing the site mean directions using the method of McFadden and McElhinny [1988]. Site 12 of normal polarity and site 14 of reversed polarity, from the Janokawauchi Formation. Equal-area projection with solid symbols on the lower hemisphere and open and dashed ones on the upper hemisphere. Circles are directions of magnetization components obtained by direct least squares line fit. Solid (dashed) lines are the best fit demagnetization great circle arcs on the lower (upper) hemisphere. Small triangles along each arc are directions used for the least squares computation. Arrows on the arcs represent the directional movement of the demagnetization paths. Stars and shaded ovals denote the site mean directions and their 95% confidence error limit.

TABLE 1. Summary of Paleomagnetic Results From the Late Miocene Uchiumigawa Group in Southeast Kyushu, Japan

Site	Lat./Lon.	S/D	N/n	(d, c)	Demag. Range	Dc	Ic	D	I	α_{95}	κ	θ	ϕ
2	31.79/131.43	056/24	7/6	(0, 7)	(<425°C)	165.3	–51.7	159.5	–28.6	15.5	23.2	77.5	47.9
3	31.60/131.44	309/08	10/7	(1,9)	225°-300° (<40 mT, <350°)	155.0	–39.3	148.6	–42.4	12.9	17.2	65.9	24.8
4	31.78/131.45	020/13	5/5	(0, 5)	(2.5-15 mT, <225°)	187.5	–48.7	174.4	–44.5	13.9	64.8	83.2	237.8
5	31.76/131.48	016/16	14/7	(14, 0)	200°-450°	135.1	–45.2	129.5	–30.9	9.6	18.2	50.8	46.0
6	31.76/131.47	000/25	7/5	(2, 5)	275°-375°, (<275°)	162.7	–46.4	143.2	–34.8	15.9	18.1	74.5	30.8
7	31.76/131.45	035/24	10/7	(2, 8)	300°-375°, (<325°)	169.1	–63.1	150.5	–43.1	9.5	30.3	74.6	101.0
8	31.76/131.46	025/19	15/8	(8, 7)	200°-450° (<12.5 mT, <250°)	143.3	–52.2	135.7	–34.8	10.6	14.3	59.1	53.4
9	31.73/131.41	015/10	8/6	(4, 4)	200°-450° (<15 mT, <325°)	132.8	–52.6	128.0	–43.6	18.8	10.2	50.5	56.6
10	31.71/131.44	355/10	9/6	(8, 1)	200°-475°, (<350°)	144.5	–49.8	135.7	–44.0	10.8	23.8	59.7	48.8
11	31.70/131.41	346/11	4/3	(3, 1)	150°-450° (150°-300°)	333.5	36.7	326.1	33.6	31.3	10.6	63.8	22.5
12	31.70/131.43	013/15	6/4	(3, 3)	5-40 mT, 150°-325° (<15 mT, <300°)	311.2	55.3	304.1	41.6	14.7	24.9	49.6	61.1
13	31.68/131.46	318/13	9/7	(2, 7)	150°-350°, (<350°)	345.6	25.7	339.2	31.1	15.7	13.2	67.6	350.8
14	31.61/131.44	340/13	9/7	(6, 3)	175°-375° (<25 mT, <325°)	132.0	–52.6	119.6	–45.2	11.6	21.4	49.8	57.0
15	31.68/131.40	353/13	13/8	(11, 2)	12.5-20 mT, 150°-400° (<350°)	146.8	–49.0	135.5	–42.1	9.6	19.8	61.5	46.6
16	31.72/131.37	312/30	11/6	(7, 4)	150°-425°, (<225°)	155.3	–45.1	122.2	–48.8	12.0	15.9	67.9	34.4
17	31.66/131.44	352/12	7/6	(6, 1)	25-40 mT, 225°-475° (<275°)	148.3	–37.0	140.3	–31.5	17.1	13.6	59.6	28.0
18	31.67/131.45	345/16	6/5	(2, 4)	225°-350° (<15 mT, <275°)	152.5	–59.3	130.7	–53.0	18.1	17.9	66.3	70.0
Mean						322.0	48.6	320.6	40.5	6.0	36.6	65.8	45.4
													A_{95} = 6.7

Lat./Lon. is latitude/east longitude; S/D is strike/dip (always clockwise from strike); N/n is number of specimens used in analysis/number of independently oriented cores; d is number of direct line-fit observations; c is number of demagnetization great circles; Demag. Range is maximum and minimun demagnetization temperature or alternating field peak intensity for least squares line and plane (in paretheses) analyses;. Dc and Ic are tilt-corrected declination and inclination; D and I are in situ declination and inclination; α_{95} and A_{95} are radius of 95% confidence circle; κ is Fisher's precision parameter; θ and ϕ are latitude and east longitude of virtual geomagnetic pole.

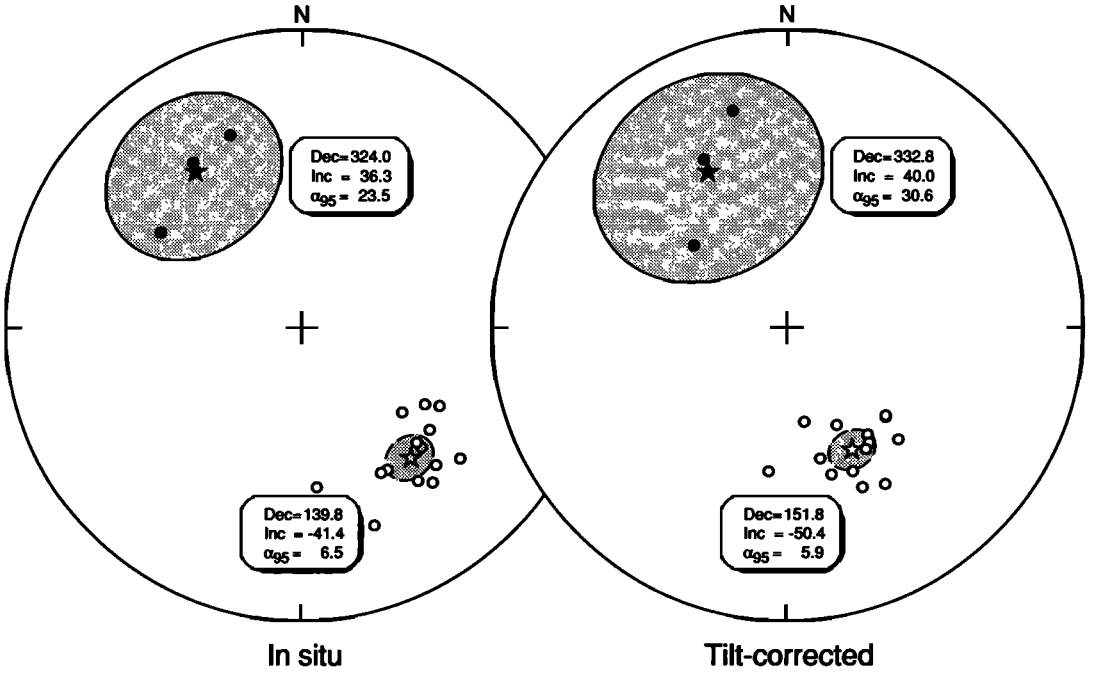


Fig. 8. Site mean directions and overall mean directions for formations with normal and reversed polarity, respectively, in geographic (left) and stratigraphic (right) coordinates. Equal-area projections with solid symbols on the lower hemisphere and open and dashed ones on the upper hemisphere. Circles denote site mean directions, and stars and shaded ovals indicate formation mean directions and the 95% errors as in attached boxes.

using only direct line data. No meaningful result was obtained for sites 1 and 19 because of poor convergence. Results from the line and plane data analysis are summarized in Table 1.

Figure 8 shows plots of the site mean directions in both geographic and stratigraphic coordinates. Converting the reversed directions to the normal through the origin, the unfolded formation mean direction for the three sites in the Aoshima Formation was D = 335.5°, I = 39.9° with α_{95} = 26.8°, and that for the 14 sites in the underlying Janokawauchi Formation was D = 331.1°, I = 50.4° with α_{95} = 6.2°. When these results are compared with one mean direction at site 16 in the Boro-ishi Formation, there is no statistically significant difference in directions of the three formations. No systematic change with time is recognized within the 14 site mean directions of the Janokawauchi Formation. It is thus reasonable to use all of these site mean directions for computing the overall formation mean direction for the entire Uchiumigawa Group.

The tilt-corrected formation mean direction for the 14 directions of reversed polarity (sites 2-10 and 14-18) was D = 151.8°, I = –50.4° with α_{95} = 5.9°, and that of normal polarity was D = 332.8°, I = 40.0° with α_{95} = 30.6°. These two mean directions are in antipodal relationship on the 95% confidence level. Inverting the reversed directions, the overall formation mean became D = 320.6°, I = 40.5°, and α_{95} = 5.9° in geographic coordinates, and D = 332.0°, I = 48.6° with α_{95} = 6.0° in stratigraphic coordinates. They both are statistically distinguishable from the present field (D = 355°, I = 45°) and the dipole field (I = 51°) directions. The mean inclinations comparable with these expected values seem to preclude the possibility of significant inclination shallowing.

No better grouping of statistical significance was achieved with correction for bedding tilt. This is principally because the layers of the Uchiumigawa Group generally dip gently to the east about a N-S strike. However, the antipodal site mean directions which definitely deflect from the present geomagnetic field suggest that these directions represent primary detrital remanent magnetizations acquired prior to bedding tilt.

Of the three sites with normal polarity, sites 11 and 12 are located in the midst of the Janokawauchi Formation of N16 stage, and site 13 belongs to the lower Aoshima Formation of N17 stage. All the others were of reversed polarity. We have only one site (site 16) in the upper Boro-ishi Formation corresponding to N15. According to a revised Neogene time scale by Berggren et al. [1985], N15 is a short stage, lasting not more than 2 m.y., within chron 11 that is equivalent to chron C5N in previous chron terminology (Figure 9). Provided that the biostratigraphic age assessment for these formations is correct, the observed reversed polarity of site 16 might be assigned to an early reversed subchron, which was correlated with a marine magnetic anomaly 5-4, according to the time scale of Harland et al. [1989]. N16 and N17 stages are much longer than N15 and have been correlated with chrons 5 to 11. Magnetic polarity frequently changed during the N16 and N17 stages. Because all the sites in the upper Janokawauchi Formation showed only reversed polarity, they probably correlate with chron 8, chron 9R, or chron 10R. Two normally magnetized horizons in the lower part could be correlated with the intervening subchrons of normal polarity. It is likely that one horizon with normal polarity in the lowermost Aoshima Formation may be linked with chron 6N or the younger part of chron 7. However, the incomplete magnetic polarity

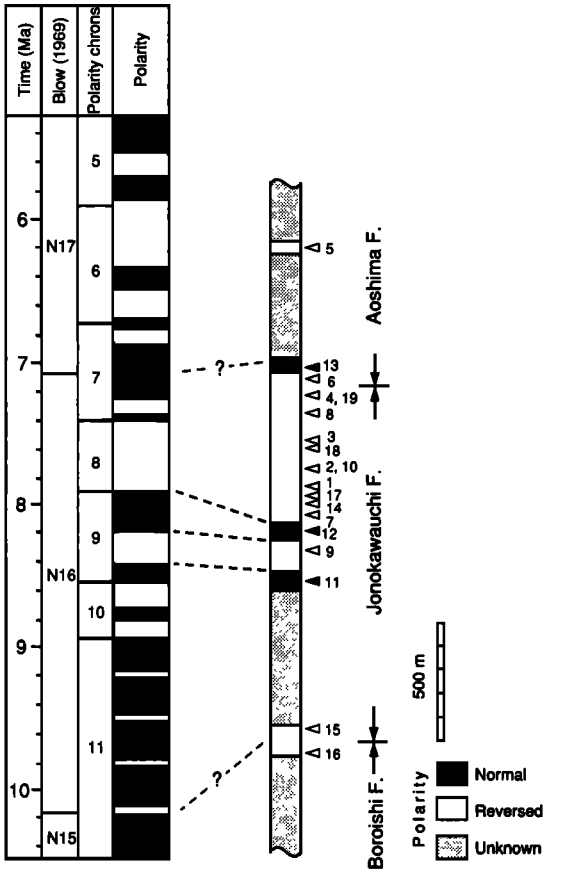


Fig. 9. Geomagnetic polarity stratigraphy of the Uchiumigawa Group, and possible correlation of the polarity zones to the late Miocene time scale (partly after Harland et al. [1989] and Berggren et al. [1985]).

observations and the lack of more detailed biostratigraphic information for the Uchiumigawa Group preclude a more precise polarity chron assignment.

DISCUSSION

CCW Rotation of South Kyushu

The mean tilt-corrected magnetization direction from the Uchiumigawa Group indicates a CCW rotation for southeast Kyushu. Because the youngest age of the rotated strata is the latest Miocene, the CCW rotation seems to have taken place during the last 6 m.y. Using the test of McFadden and Jones [1981], the paleomagnetic direction of the Uchiumigawa Group (D = 322.0°, I = 48.6° with α_{95} = 6.0°) is in statistical agreement with that for Tanegashima Island of the northern Ryukyu arc (D = 331.2°, I = 41.3° with α_{95} = 9.9° [Kodama et al., 1991]). This means that they share a common mean direction that can be computed using all the site mean directions of these two regions. The common direction becomes D = 333.2°, I = 45.1° with α_{95} = 4.9°, which deviates from the dipole field direction by 26.8° ± 5.6° in declination and 0.1° ± 3.9° in inclination (the errors are of 95% confidence limits computed using correction factor of Demarest [1983]). The difference in declination is significant, but that in inclination is not. The Eurasian mean paleopole has been in reasonable agreement with the geographic pole from the late Miocene onward [Irving and Irving, 1982]. No relative motion has been recognized between Eurasia and southwest Honshu since the middle Miocene [e.g., Otofuji et al., 1991]. Therefore it is concluded that both south Kyushu and the northernmost part of the Ryukyu arc underwent 27° ± 6° of CCW tectonic rotation relative to the Eurasian continent after the latest Miocene, without latitudinal translation. Several westerly deflected magnetization directions were also obtained from the middle Miocene granite in south Kyushu (2 in Figure 2) by Ishikawa and Torii [1986], who estimated a 45° ± 24° westward shift.

The pre-Miocene strata in southwest Kyushu exhibit a distinct southward bend from the typical ENE-WSW strike of southwest Japan (Figures 2 and 3). This change in structure has been named the Hokusatsu Bend [Hashimoto, 1962], the Hitoyoshi Bend [Murata, 1987], or the Nojiri Bend [Murata, 1987] after each locality. These bends caused conical folds of the pre-Miocene strata [e.g., Fabbri et al., 1987]. To explain these bends, Murata [1987] proposed a model where both southwest Honshu and north Kyushu had rotated CW relative to south Kyushu and the north Ryukyu arc. Faure et al. [1988] also supported this rotation on the basis of the discrepancy in the lineation trend of metamorphic rocks between west Kyushu and southwest Japan. All of them interpreted that the bend formed synchronously with the Japan Sea opening at the middle Miocene, with south Kyushu fixed. Their model assumes no relative motion between south Kyushu and Eurasia since the middle Miocene. However, their interpretation is incompatible with paleomagnetic data from the Miocene rocks in south Kyushu [Kodama et al., 1991; this study]. We emphasize that the amount of the bend (about 30°) coincides well with a proposed CCW rotation angle of south Kyushu, 27° ± 6°. According to recent well-dated paleomagnetic studies [e.g., Hayashida et al., 1991], the CW rotation of southwest Japan occurred most likely between 16 and 14 Ma, or between N8 and N10 stages. In contrast, the middle to late Miocene paleomagnetic data from south Kyushu exhibit consistent, westward declinations from N10 through N17 stages. This means that the CCW rotation postdates the CW rotation and the Japan Sea opening. Thus the structural bend in southwest Kyushu was caused after the latest Miocene by the CCW rotation irrespective of these older tectonic events. It follows that Honshu and Kyushu had kept a more linear configuration than at present until the latest Miocene.

The Uchiumigawa Group is overlain to the north by the latest Miocene to Pliocene Miyazaki Group (Figure 3). The upper portions of the Miyazaki Group strike NNE-SSW, while the southerly portions of the group are bent sharply westwards by several sinistral strike-slip faults with EW strike [Suzuki, 1987]. To the south of these faults, the Paleogene basement is exposed locally, with the same N-S strike as the older layers south of the Nojiri Bend. The fault displacement is reflected by the trend of coastline which shifts about 10 km seaward at the south of the faulted zone (Figure 3). These faults extend to the Nojiri and Hokusatsu bends to the west. There is no major fault cutting across the strike of the Neogene and Paleogene formations to the north of the Miyazaki Group. Paleomagnetic results from the middle Miocene granite in the Sobosan-Okueyama area (7 in Figure 2) suggest no tectonic rotation. We thus speculate that the faulted zone in the Miyazaki Group

and its western extension to the Hokusatsu Bend may be the possible tectonic northern boundary of the rotated south Kyushu block.

CCW Rotation in Extensional Tectonic Regime

Detailed bathymetric and seismic studies [Letouzey and Kimura, 1986; Sibuet et al., 1987] suggest that the southern and middle parts of the Okinawa trough are in an incipient spreading stage of back-arc basin evolution, while the northern part is only in an early rifting stage. To account for the asymmetric extension in the Okinawa trough, Viallon et al. [1986] proposed a numerical model on the basis of a retreating trench hypothesis. Their model assumes two anchor points; one due to the collision in Taiwan and the other to the buoyant subduction of the Kyushu-Palau ridge beneath the southwest Japan arc. They predicted a CW rotation for the south Ryukyu arc and a smaller CCW rotation for the north Ryukyu arc. The CW rotation was detected by a paleomagnetic study on the south Ryukyu arc [Miki et al., 1990] (Figure 10). However, the expected CCW rotation for the north Ryukyu arc is not more than 10°; that amount of rotation is smaller than the paleomagnetic estimate in this study (27° ± 6°). It has been proposed that the rotation of an island arc is caused by a wedge-shaped spreading of back-arc oceanic crust [e.g., Otofuji and Matsuda, 1987]. However, because of the absence of back-arc oceanic crust, no significant rotation is expected for south Kyushu and the north Ryukyu arc. These discrepancies between the observed and predicted paleomagnetic directions demonstrate that these models are not completely satisfactory. This suggests that some other process must be affecting the CCW tectonic rotation.

Central Kyushu is characterized by a series of volcanic depressions trending ENE-WSW [e.g., Kamata, 1989] (Figures 2 and 10). Many active faults responsible for their formation have been detected by submarine sonic surveys [e.g., Shimazaki et al., 1986] as well as on-land geology [e.g., Ikeda, 1979]. Radiometric age determinations [e.g., Kamata and

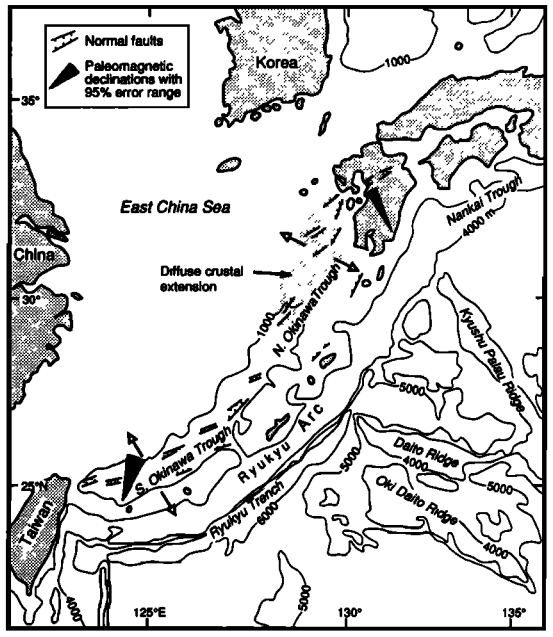


Fig. 10. General morphotectonic map of Japanese arc, Ryukyu arc, Okinawa trough, and the vicinity, with paleomagnetic data for the southern Okinawa trough [Miki et al., 1990] and south Kyushu (this study) including the northern Okinawa trough.

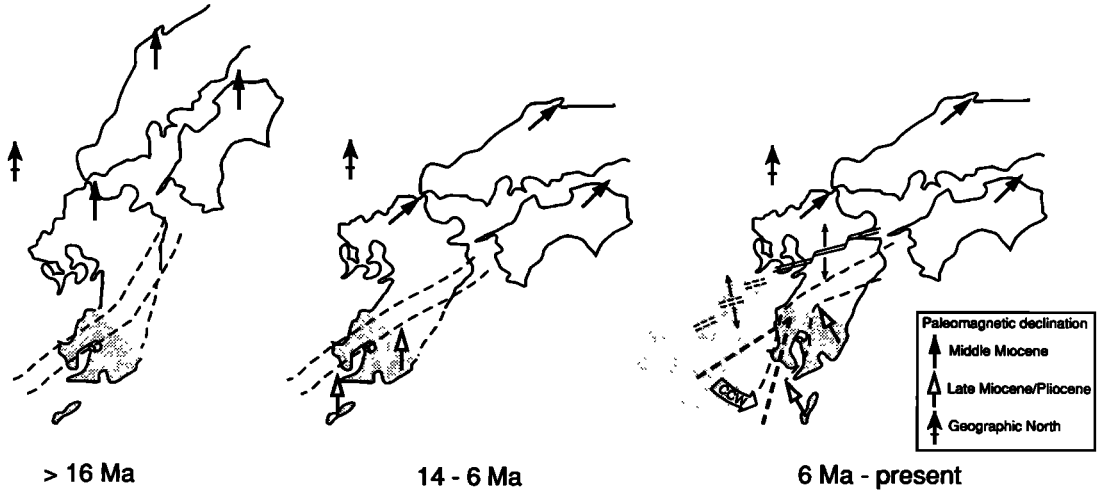


Fig. 11. Tectonic development of southwest Japan including southwest Honshu, Shikoku, and Kyushu since the middle Miocene. Arrows denote orientations of the representative areas based upon paleomagnetic studies. The shaded area is a south Kyushu block that rotated counterclockwise (CCW) during the last 6 m.y. Dotted lines show the general strike of the pre-Miocene strata in the outer zone of SW Japan. (Left) Prior to a clockwise (CW) rotation of SW Japan at 14-16 Ma. (Middle) After the CW rotation, assuming north Kyushu rotated CW along with SW Honshu and Shikoku. The middle Miocene Kukinaga Group in Tanegashima Island and the late Miocene Uchiumigawa Group in SE Kyushu were deposited after the CW rotation and magnetized to the N-S direction. (Right) The latest Miocene to the present, during when both south Kyushu and a part of the north Ryukyu arc rotated CCW relative to the others. The CCW rotation had the pre-Miocene strata in SW Kyushu bent to the south by approximately 30°.

Watanabe, 1985] showed that the central graben has evolved since 5 Ma, along with coeval volcanism. Geodetic surveys [Tada, 1985] prove that the N-S crustal extension is currently under way in central Kyushu. Focal mechanism solutions for earthquakes also suggest the N-S extension [Yamashina and Mitsunami, 1977].

All these geological and geophysical studies suggest that central Kyushu has been subjected to a N-S crustal stretching under an extensional tectonic regime since the early Pliocene. According to our paleomagnetic data in this study, the southern half of Kyushu has rotated CCW during this period. Figure 11 illustrates a model of the tectonic evolution of southwest Japan since the early middle Miocene. A problem may arise in this model; what filled the vacancy behind south Kyushu that had been produced by the post-late Miocene CCW rotation? There is no evidence for back-arc spreading in the area west of central Kyushu, except for several submarine normal faults suggesting crustal rifting (Figure 10). One possible answer could be widespread rheological stretching of the crust in this area, which had been softened by increasing heat supply from the asthenosphere. This interpretation supports the views of Faure and Charvet [1991] that some marginal seas in east Asia, such as the Okhotsk and East China seas, are underlain by stretched continental crust. The rotation under extension is analogous to rotations of extended terrains in the Basin and Range province [e.g., Janecke et al., 1991]. This similarity suggests that vertical-axis rotation may generally occur in extensional regions.

The rotation of south Kyushu is in remarkable contrast with that of central Honshu [e.g., Hyodo and Niitsuma, 1986], where compression has been dominant due to the collision of the Izu-Bonin arc with the Honshu arc. Kodama et al. [1991] suggested that the CCW rotation of Tanegashima Island had been caused by the collision of the Kyushu-Palau ridge with the southwest Japan arc. They proposed that the pre-Miocene strata in south Kyushu had been bent to form a cusp under the compression derived from the ridge indentation into the arc, as observed at present in central Honshu. Because of the limited time range of the rotated strata in Tanegashima Island, they concluded that the rotation had taken place not earlier than the late Miocene. The present study unraveled that the rotation had taken place much later than ever expected, that is, only for the last 6 m.y. The strain field in central Kyushu during most of this period seems to have been characterized by a N-S stretching rather than shortening. We thus believe that the collision, or the buoyant subduction, of the Kyushu-Palau ridge has never played an important role for the south Kyushu deformation since 6 Ma. It is unlikely that either the ridge indentation or the strong coupling with the overriding plate capable of producing compressional field has been present for this period.

CONCLUSIONS

Paleomagnetism of sedimentary rocks of the late Miocene Uchiumigawa Group in southeast Kyushu indicates 27° ± 6° of counterclockwise rotation about a vertical axis. This rotation is in agreement with that previously obtained from the middle Miocene deposits exposed in Tanegashima Island of the northern Ryukyu arc. This indicates that both south Kyushu and the northern Ryukyu arc rotated counterclockwise with respect to southwest Japan after the latest Miocene, for the last 6 m.y. This intra-arc rotation yielded a cusplike bend of the pre-Miocene strata in southwest Kyushu. This result opposes a previous model which interprets the bend as a consequence of the middle Miocene clockwise rotation of southwest Japan.

An extensional tectonic regime has been predominant for central Kyushu from at least 5 Ma to the present. It is thus unlikely that either the collision or the buoyant subduction of the Kyushu-Palau ridge produced this intra-arc deformation, as seen in the central Honshu-Izu arc collision area. Since the back-arc spreading is only at an incipient rifting stage behind the northern Ryukyu arc, the conventional model which ties arc rotation to back-arc basin formation cannot fully account for the rotation of south Kyushu. We propose that widespread stretching of continental crust prior to the back-arc opening is able to cause an arc to rotate.

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